Lecture-#10

Natural Language Processing

A8706

Syntax Analysis

Probabilistic Context-Free Grammars:

1. Define Probabilistic context-free grammar?

A probabilistic context-free grammar is the same as context-free grammar, but here we have probabilities assigned to each rule in the grammar.

2. What is the significance of the parameters in PCFG?

The parameter $q(\alpha \rightarrow \beta)$ can be interpreted as the conditional probability of choosing rule $\alpha \rightarrow \beta$ in a left-most derivation, given that the nonterminal being expanded is α.

3. Can we apply the CKY algorithm to the Probabilistic Context-Free Grammar?

Yes, the Cocke-Younger-Kasami algorithm applies to a restricted type of PCFG, which is in Chomsky standard form (CNF).

Probabilistic Context-Free Grammars:

4. What is probabilistic parsing?

Probabilistic parsing uses dynamic programming to compute the most likely parse of a given sentence.

5. Why do we use the CYK algorithm?

We use the CYK algorithm to find whether the given string belongs to a language of grammar or not.

Probabilistic Context-Free Grammars:

$N = \{\text{S, NP, VP, PP, DT, Vi, Vt, NN, IN}\}$
$S = \text{S}$
$\Sigma = \{\text{sleeps, saw, man, woman, dog, telescope, the, with, in}\}$

$R, q =$

S	$\rightarrow$	NP	VP	1.0
VP	$\rightarrow$	Vi		0.3
VP	$\rightarrow$	Vt	NP	0.5
VP	$\rightarrow$	VP	PP	0.2
NP	$\rightarrow$	DT	NN	0.8
NP	$\rightarrow$	NP	PP	0.2
PP	$\rightarrow$	IN	NP	1.0

Vi	$\rightarrow$	sleeps	1.0
Vt	$\rightarrow$	saw	1.0
NN	$\rightarrow$	man	0.1
NN	$\rightarrow$	woman	0.1
NN	$\rightarrow$	telescope	0.3
NN	$\rightarrow$	dog	0.5
DT	$\rightarrow$	the	1.0
IN	$\rightarrow$	with	0.6
IN	$\rightarrow$	in	0.4

Probabilistic Context-Free Grammars:

- The maximum-likelihood parameter estimates are

$$q_{ML}(\alpha \rightarrow \beta) = \frac{\text{Count}(\alpha \rightarrow \beta)}{\text{Count}(\alpha)}$$

  where $\text{Count}(\alpha \rightarrow \beta)$ is the number of times that the rule $\alpha \rightarrow \beta$ is seen in the trees $t_1 \ldots t_m$, and $\text{Count}(\alpha)$ is the number of times the non-terminal α is seen in the trees $t_1 \ldots t_m$.

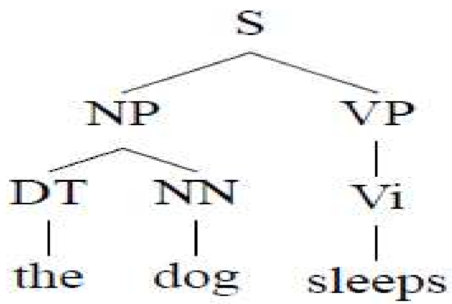
For example, if the rule VP $\rightarrow$ Vt NP is seen 105 times in our corpus, and the non-terminal VP is seen 1000 times, then

$$q(\text{VP} \rightarrow \text{Vt NP}) = \frac{105}{1000}$$

Probabilistic Context-Free Grammars:

$$
\begin{aligned}
\sum_{\alpha \rightarrow \beta \in R : \alpha = \text{VP}} q(\alpha \rightarrow \beta) &= q(\text{VP} \rightarrow \text{Vi}) + q(\text{VP} \rightarrow \text{Vt NP}) + q(\text{VP} \rightarrow \text{VP PP}) \\
&= 0.3 + 0.5 + 0.2 \\
&= 1.0
\end{aligned}
$$

To calculate the probability of any parse tree t, we simply multiply together the q values for the context-free rules that it contains. For example, if our parse tree t is



then we have

$$
\begin{aligned}
p(t) = \; & q(\text{S} \rightarrow \text{NP VP}) \times q(\text{NP} \rightarrow \text{DT NN}) \times q(\text{DT} \rightarrow \text{the}) \times q(\text{NN} \rightarrow \text{dog}) \times \\
& q(\text{VP} \rightarrow \text{Vi}) \times q(\text{Vi} \rightarrow \text{sleeps})
\end{aligned}
$$

Models for Ambiguity Resolution in Parsing

Generative models for parsing :

❖Generative models for parsing are a class of models that focus on generating the most likely parse tree for a given sentence.

❖They are often probabilistic models that assign probabilities to different parse trees and aim to find the highest probability parse.

❖**Probabilistic Context-Free Grammars (PCFGs):**

PCFGs are a classic generative model for parsing. They extend context-free grammars by associating probabilities with each production rule. The probability of a parse tree is calculated as the product of the probabilities of its constituent rules. Parsing involves finding the parse tree with the highest probability.

❖**Lexicalized PCFGs (L-PCFGs):**

L-PCFGs enhance PCFGs by incorporating information about the words in the sentence. Instead of just relying on syntactic structure, L-PCFGs consider the words and their relationships when assigning probabilities to parse trees. This makes them more sensitive to word-specific and syntactic ambiguities.

Models for Ambiguity Resolution in Parsing

Generative models for parsing :

❖**Transformation-Based Models:** Transformation-based models, such as the transformation-based learning (TBL) algorithm, are generative models that iteratively transform one parse tree into another. These models learn a sequence of operations to resolve parsing ambiguities based on training data.

❖**Tree Substitution Grammar (TSG):** TSG is a generative model that generates parse trees by iteratively replacing tree fragments with other trees. TSG can capture various tree structures and is useful for resolving structural ambiguities in parsing.

❖**Bayesian Grammar Models:** Bayesian grammar models apply Bayesian inference to probabilistic grammars, allowing for uncertainty modeling and ambiguity resolution. These models can provide posterior probabilities for different parse trees, which can be useful in situations where multiple parses have similar probabilities.

Models for Ambiguity Resolution in Parsing

❖Discriminative models for parsing :

❖Discriminative models for parsing are a class of models that focus on directly predicting the most likely parse or parsing decisions for a given input sentence without explicitly generating alternative parse trees.

❖ These models are often trained to maximize the likelihood of the correct parse, making them well-suited for structured prediction tasks.

❖**Conditional Random Fields (CRFs):** Conditional Random Fields are a widely used discriminative modeling technique for various structured prediction tasks, including parsing. In the context of parsing, the input sentence's features (e.g., words, part-of-speech tags) are used to conditionally model the probability of different parse tree structures. CRFs can handle both syntactic and semantic parsing tasks and are known for their strong performance.

Models for Ambiguity Resolution in Parsing

❖Discriminative models for parsing :

❖**Maximum Entropy Models (MaxEnt):** Maximum entropy models, often used for part-of-speech tagging and syntactic parsing, assign probabilities to potential parses based on a combination of features derived from the input sentence. These features may include word identities, part-of-speech tags, and contextual information. MaxEnt models aim to maximize the likelihood of the correct parse based on the given features.

❖**Structured Support Vector Machines (SVMs):** Structured Support Vector Machines are used for structured prediction problems, including parsing. These models seek to find a hyperplane that best separates the correct parse from other possible parses in a high-dimensional feature space. SVMs are effective in learning complex decision boundaries for parsing tasks.

❖**Transition-Based Parsing Models:** Transition-based parsers are discriminative models that predict parsing actions to build a parse tree incrementally. These models use a set of parsing actions and associated classifiers to make decisions at each step. They are particularly useful for dependency parsing.


Thank you!!!
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